



On the optimal integration of legacy and SDN communication nodes in emergency scenarios

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Accepted: 21 November 2024 / Published online: 17 December 2024
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Abstract

Spontaneous networks may need to establish connections with already-in-place devices and their resources, irrespectively of their capabilities, for example, in emergency scenarios. Interoperability and integration are a key issue when combining multiple wireless networks and heterogeneous devices. To address this issue, this article extends the SDN concept, and presents a new hybrid network architecture that combines legacy devices in a wireless data plane, therefore increasing the range, and making the network more efficient, dynamic and programmable. In this article, we propose an optimized adaptive routing approach in a network architecture that joins SDN-capable devices with legacy devices for emergency scenarios. This solution takes into consideration the impact that a flow can have in the network, therefore minimizing its channel interference, and to a certain degree, balancing the load in the network's devices. In this work, the importance of the network and service parameters is optimized through machine learning for each flow's data rate, and then feed into the routing approach to determine the best hybrid legacy and SDN-based path. Results show that the new architecture can successfully integrate the legacy devices in the wireless software defined data plane, and utilize them to increase the performance. The combination of delivery ratio and delay are always relevant to every data rate. The link stability is more relevant for lower data rates, while the break probability is more relevant for higher data rates. Moreover, the number of hops and possible interference combined have a high importance in every data rates, and the nodes' speed knowledge can have a good impact on the network performance.

Keywords Legacy · SDN · Path selection · Hybrid wireless software-defined networks · Data rate based optimization

1 Introduction

In environments like surveillance, Wi-Fi access, emergency scenarios, large-scale events, and others, the communication network needs to be scalable, autonomous, flexible,

self-reconfigurable, adaptive, and self-organizing, to deal with the unknown heterogeneity and dynamics of the environment. To meet these requirements, spontaneous networks can be used Deepak G.C. et al. [12], where the network is temporarily created, replaced, or extended with the already existing devices to enhance the communication among users Evans et al. [15].

In these environments, the already in-place devices cannot be modified, therefore creating several problems like heterogeneity, compatibility, lack of prior knowledge of the services available in each of the networks, or the lack of best overall routing path.

The Software Defined Network (SDN) paradigm can be used to solve the last problem by gaining global knowledge of the state, available topology, and services' requirements, providing better Quality of Service (QoS) Bellavista et al. [6], making these networks more efficient and flexible.

However, the integration of SDN in wireless environments remains a challenge Liu et al. [28], due to their weak

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wireless interface support, no support for legacy devices, wired-based protocols, etc.. Despite this, the community believes that the implementation of an SDN will bring benefits to such networks Kafetzis et al. [21], even if not supported by all the network nodes.

Nevertheless, for example, in an emergency scenario, there is a need where all nodes, even those not supporting the technology, to form one network in order to increase its scale and, therefore, the services provided. Therefore, we consider a hybrid network the one where non-SDN capable devices, also known as legacy devices, can communicate and form a network with the SDN capable devices, taking advantage not only of the legacy devices capabilities, but also the SDN capable devices, increasing the range, programmability, flexibility, and network capabilities. The connection of out-of-the-box legacy devices to the Software Defined Wireless Network (SDWN) devices in a hybrid network is quite challenging, and several network services need to be adapted and supported Amin et al. [2]. One of those services is the routing protocol in both SDN and legacy devices, therefore creating a hybrid network with hybrid devices Sandhya et al. [33].

With this new architecture, we also need network management that can cope with the heterogeneity.

Traditionally, network management relies on static configurations and manual adjustments, which makes it challenging to respond swiftly to changing network conditions or to accommodate diverse applications and services. The discovery mechanisms in SDN are designed to identify, classify, and expose network resources, ensuring that controllers have real-time access to vital information about the network's topology, available bandwidth, device capabilities, and more. However, these mechanisms need to be adapted to support wireless networks Chen et al. [8].

QoS has long been an important topic in networking, ensuring that data and applications are delivered with the requisite performance and reliability. QoS-sensitive or QoS-aware routing attempt to improve the network utilization by using paths that traditional path selection mechanisms would not Apostolopoulos et al. [3]. SDN bring the ability to adjust QoS mechanisms and their routing metrics in real-time according to the network state, which can increase the QoS performance Qureshi et al. [31]. However, many traditional routing metrics rely on uniform policies, where all network traffic is treated equally, allocating resources based on general parameters Vu Khanh et al. [39]. This one-size-fits-all approach falls short in addressing the nuances of modern network demands.

Service optimization, on the other hand, tailors QoS policies to individual applications and services, acknowledging that not all data is created equally. It recognizes that a real-time video call requires a different network treatment from a file download or e-mail transfer. The

significance of service-based optimization is underscored by its ability to offer several key advantages. By recognizing the distinct requirements of various services and applications, SDN have the capability to dynamically allocate resources more efficiently, prioritizing critical data and ensuring low-latency communication for delay-sensitive services.

In this article, we propose a solution for the challenges described by Amin et al. [2]: how to integrate legacy and SDN nodes in emergency scenarios with an end-to-end simple network management, path formation, support for wireless environments in out-of-band controllers, and handling the channel interference. We consider a network architecture that connects SDN-capable devices with legacy devices, in which, by integrating a legacy routing mechanism in the SDN-capable devices, we can extend the range of applications given by the SDN. Moreover, by allocating different wireless technologies in each plane (the control-plane having higher range), no interference from the control messages can be felt in the data-plane. We also propose a routing algorithm and a discovery mechanism that can take advantage of all the network resources, using both legacy and SDN devices, making their network performance more efficient than traditional legacy networks. Moreover, our solution takes into consideration several aspects of the network when calculating a new path. One of them is the impact that a flow can have in the network, therefore minimizing its channel interference, and to a certain degree, balancing the load in the network's devices. In this work, the importance of the network and service parameters is optimized through machine learning for each flow's data rate, and then feed into the routing approach to determine the best hybrid legacy and SDN-based path.

As a summary, this work presents the following main contributions:

- The approach for a network architecture made up of both legacy and SDN-capable devices;
- The integration of network interfaces of different wireless technologies in an SDN network;
- A monitoring tool capable of gathering information about communication links and Hybrid SDN (H-SDN) devices by using the legacy topology discovery mechanism;
- The research of an optimized service-based path search routing mechanism, which integrates both legacy and SDN devices;
- A machine learning based approach to optimize the importance of the network and service parameters for each flow's data rate, which will then feed the service-based path search routing mechanism;
- Performance evaluation of the proposed routing mechanism in a simulation environment considering mobile scenarios in a hybrid network.

The remaining of this article is organized as follows. The related work is discussed in Sect. 2. The proposed approach for a spontaneous wireless Hybrid SDN is addressed in Sect. 3. The proposed method for calculating the best path is described in Sect. 4. The simulation details, along with the evaluation setup, as well as the scenarios and the results, are presented and discussed in Sect. 5. Finally, Sect. 6 concludes the article and discusses future research directions.

2 Related work

This section identifies relevant related work on Hybrid SDN and an adaptive heuristic functions for path searching.

As stated before, SDN can bring benefits to wireless networks Zahmatkesh and Kunz [42]. The networks benefit from management unification, flexible control, spontaneous and dynamic deployment, efficiency management, and QoS improvement Qureshi et al. [31]. However, standard protocols and devices are only prepared for wired connections, and adapting such protocols to wireless environments brings many challenges, like device heterogeneity, topological and interface diversity, protocol integration, and node mobility Latif et al. [26].

According to a comparison study Haji et al. [18], this architecture can be more efficient than traditional/legacy architectures. Moreover, by integrating an SDN architecture with the existing legacy networks, advantages like higher coverage, less management cost, and cheaper networks can be achieved while providing the same services Galán-Jiménez [16]; Vissicchio et al. [38].

QoS-oriented services can increase the performance in such a novel network. Moreover, the SDN paradigm makes it easier to create a QoS-aware routing protocol since a global view is provided in a centralized manner Han et al. [19]. One of the key issues of such protocols is to identify a suitable route that satisfies multiple criteria while providing efficient network resource utilization Korkmaz and Krunz [24]. This implies the need for an optimality requirement in a path's feasibility evaluation, like hop count or weight Korkmaz and Krunz [24].

Guo et al. [17] describe a hybrid network in a Wide Area Network, where hybrid SDN devices are among legacy switches, utilizing the same legacy routing protocol to communicate with them. The hybrid SDN device uses both the legacy routing mechanism and the OpenFlow controller, which only forwards flows when they impact the load-balancing performance of the network. The controller utilizes a routing metric based on the edge's load and the total link load in the network to avoid long paths that increase the traffic load on the entire network.

However, the improvement of additional network parameters in a hybrid heterogeneous network requires consideration of various metrics, despite the increased complexity associated with a higher number of routing parameters Kaur and Kumar [22]. Heuristic functions or approximation algorithms are often used to deal with this Elkael et al. [14].

The global view and centralized control of a SDN allows for the development of a QoS-aware routing algorithm with the current status of links in its design. The statuses are indispensable in wireless networks, and their use can lead to a better performance Han et al. [19]; Kumar et al. [25]. However, studies still prioritize the hop count over QoS for path search Yang et al. [41], and fewer studies use legacy devices in their SDN.

Chen et al. [9] presented a multi-constraint shortest path algorithm in SDN for UAVs based on bandwidth, delay, and data stream size. The algorithm is tested on Mininet, where the transmission rate increases in 16% while the jitter is controlled at 0.2ms when compared to a traditional network. Although it is implemented in SDN, it does not take into consideration the state of the network for wireless challenges.

Kumar et al. [25] presented a centralized QoS-Aware Routing algorithm in SDN where the Dijkstra's algorithm is used to find the best path. The algorithm is tested in the Network Simulator 3 (ns-3) in which the computation time is reduced significantly. However, it only takes into consideration the congestion when computing the link cost, and it is not applied to wireless SDNs.

Hua et al. [20] described a weighted-sum based on the distance and cost criteria, for a multi-criteria heuristic function used in the Dijkstra path planning algorithm, which can identify the best path without the involvement of all alternatives. This approach, when compared to all the presented alternatives in Mohammadi and Hunter [30], has the best performance Alwan [1].

Beshley et al. [7] developed an adaptive flow routing model in SDN, which uses link activation, re-utilization, and link-utilization costs to find the best path. The objective is to develop a model for effective routing with high degree of flexibility by using a general routing metric composed of a set of weighting coefficients. The work tries to maximize the number of flows, while minimizing the cost. However, the flow characterization is inaccurate and some flows re-utilize rules for other flows without calculating their cost.

Better parameters for flow differentiation need to be used. For example, if a flow has different data rates, the cost to the network will be different; parameters that are not important for a low data rate flow can be important to high data rate flows. Thus, we propose an adaptive multi-criteria heuristic to find the best path depending on the data rate of the flow, and considering both SDN and legacy devices in the path. To the best of our knowledge, there is no description on what

exact parameters are more important for each data rate, but only heuristics that prove to have higher performance for a specific problem. Thus, to find the most important paths in a hybrid SDWN with heterogeneous legacy devices, we optimize the importance of each parameter based on the network's delivery and delay, therefore maximizing the performance and utilization of its resources through the different types of nodes.

3 Hybrid wireless software defined network architecture

SDWN is a recent approach to manage and optimize wireless networks using the SDN paradigm Assegie and Nair [4]. SDWN can bring the advantages of SDN to wireless networks, providing greater flexibility, scalability, simplified network management, real-time programmability, and cost-savings for devices. Moreover, with a global network view, changes in the network can be known faster, ultimately improving its performance.

The architecture considered in this work comprises the use of already-in-place network devices, which can be SDN-capable or legacy (if they do not support SDN). To address the issue of heterogeneity in similar problems, common solutions either implement a hybrid SDN, or detach the SDN management mechanism. Both approaches have their own advantages and disadvantages. In the former, monitoring tools, routing protocols, and other applications will have to work simultaneously on both legacy and SDN. Because of that, if most of the network comprises legacy devices that do not support the SDN protocols, then no benefit is achieved. In the latter case, the mechanisms developed for standard SDN networks will have to be re-implemented with a problem-specific solution. With few legacy devices, this is often a more time-consuming approach than a hybrid SDN.

SDN has three planes: the Data Plane, where all the wireless network devices operate; the Control Plane, where the network configuration, monitoring, and troubleshooting of the network is done; and the Application Plane, where network applications run. Between these planes, two connecting protocols exist: Northbound Interface (between the application and Control planes); and Southbound Interface (between the Data and Control Planes), where protocols like Openflow (OPF), ForCES, P4, NetConf, or others, are used.

Our proposed solution follows a hybrid SDN model, because not all of the network will be SDN-capable. Therefore, the following must be implemented: changes in the routing protocol to accommodate for routing in wireless environments; a southbound interface support for wireless connections; and a monitoring tool working on both legacy and SDN-capable devices. The routing protocol will be later described in Sect. 4.

3.1 Wireless openflow interfaces

One of the most used Southbound interfaces is OpenFlow. It is a communication protocol that connects the data plane devices to the controller. Each device interface is mapped as an OpenFlow port, where each received packet goes through the OpenFlow tables, which are filled by the controller. Current OpenFlow software only supports MAC address scheming based on IEEE 802.3 standard (wired networks), so neither the 802.11 frame can be processed by OpenFlow, nor can a 802.3 frame be used over the wired link.

However, there have also been efforts to extend its principles to wireless networks. Sharma et al. [34] proposed the use of tunneling, but this approach leads to network overhead, a decrease of initial performance, and an increase in complexity. Rademacher et al. [32] use *4addr* mode, which may not be supported by all interfaces. Detti et al. [13] described a packet header modification with an extra module for MAC rewriting, which is better suited for our proposed architecture.

A header modification module enables off-the-shelf Controllers and OpenFlow solutions to readily work with the new wireless devices. In the proposed architecture, when a packet is received, the wireless interface is mapped to a wireless OpenFlow port. However, OpenFlow ports are only expecting IEEE 802.3 header. Therefore, we proposed in Sousa et al. [36] a header modification module where the IEEE 802.11 header is substituted by a IEEE 802.3 header. The receiver and transmitter addresses of a IEEE 802.11 header are mapped to IEEE 802.3 source and destination addresses respectively, and can then be interpreted by the OpenFlow protocol. Afterwards, the inverse occurs before forwarding the packet through the forwarding interface.

Using a common platform and standard simplifies the process of extending different technologies. This medium-independent solution is more efficient than solutions where medium-specific extensions are proposed Taniuchi et al. [37].

When a packet's header is altered Sousa et al. [35], there is a possibility that the sender's MAC address may not match the sender's IP address. As a result, the mechanism used to discover the network topology needs to be adjusted, which is described in Sect. 4. If these mechanisms are not working correctly, the routing logic is affected, which can lead to decreased performance Azzouni et al. [5].

3.2 Proactive topology management

In a SDWN environment, topology discovery mechanisms are crucial to maintain an up-to-date and accurate understanding of the network's physical and logical layout Khan et al [23]. These mechanisms give the centralized controller real-time information about the network's topology, which

is essential for making informed routing decisions and optimizing resource allocation.

As stated before, our scenario is comprised of legacy and SDN devices. To connect both, without changing the legacy devices, the SDN devices need to become hybrid and run a standard legacy routing protocol that has a local routing table and discovery mechanism. In the proposed solution, the OLSR Clausen and Jacquet [11] is used as the protocol, as is one of the most used routing protocols for ad-hoc legacy networks. Our proposed solution is prepared to handle proactive legacy routing protocols, like OLSR, that save the information about all the nodes in the network in a local table. However, for other protocols, the application in the controller would also need to adapt to be able to interpret these legacy routing messages.

Having two routing protocols running, each having its own discovery mechanisms, will increase the overhead in the network. Additionally, OpenFlow's topology discovery mechanism (LLDP) was designed primarily for wired connections, so it would not work on wireless networks. To minimize the additional overhead, LLDP was replaced by a new topology discovery mechanism that utilizes the messages already sent periodically by the legacy discovery mechanism of the OLSR protocol. These OLSR messages ensure the existence of a local updated neighbors table, which is then used to compute the topology message.

Our proposed topology discovery mechanism works in both passive and active modes. In active mode, the controller can instruct an SDN-capable device to send messages to gather local statistics. As shown in Fig. 1, the hybrid SDN device builds a topology message that enables the controller to maintain an up-to-date view of the network. This message is generated proactively-both periodically and upon changes in connections-and includes topology information from the OLSR table, as well as local statistics collected when a packet is received.

Furthermore, between periodic transmissions, if a neighbor is added or removed, the mechanism is triggered to send a message with this new information, reducing the time required to recover from changes in the topology.

This mechanism ensures an up-to-date view of the data plane in the controller, with both information from the legacy and hybrid SDN devices.

4 Hybrid routing protocol

When dealing with a spontaneous network, some traffic may have specific requirements that could eventually affect the survivability and performance of the network. To determine the best path in a hybrid SDN and non-SDN network, these constraints and requirements must be taken into account.

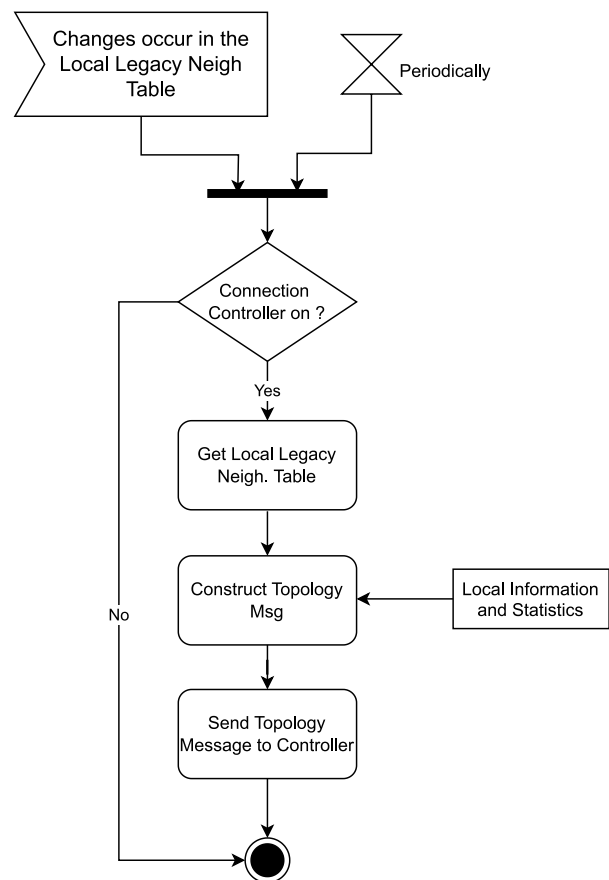


Fig. 1 Proactive Topology Management in the hybrid SDN device. The topology messages are constructed periodically and when changes exist in connections. These messages use the information gathered when a packet is received

Computing a path with more than one additive constraint is NP-hard Wei et al [40]. Hence, numerous researchers have shifted their focus towards heuristic functions or approximation algorithms to tackle this problem Wei et al [40]. Many heuristic functions only consider the network's state, which may not be enough. To address this, we proposed an Adaptive QoS-aware optimized route path calculation, in which we utilize Dijkstra with a heuristic function composed of several service-dependent parameters and weights. Instead of calculating the best path based on some parameters, we integrate all the parameters available for the controller, and use an optimization algorithm to find the best parameters for specific Data Rates, providing a better resource utilization, and increasing performance.

As shown in Fig. 2, our proposed Routing Algorithm comprises both SDN and legacy routing mechanisms, making it a Hybrid Routing Algorithm. In a mobile scenario, the connection to the controller may not always be available. Therefore, when no controller is in reach, the hybrid device behaves like a legacy device, thus making it fault-tolerant.

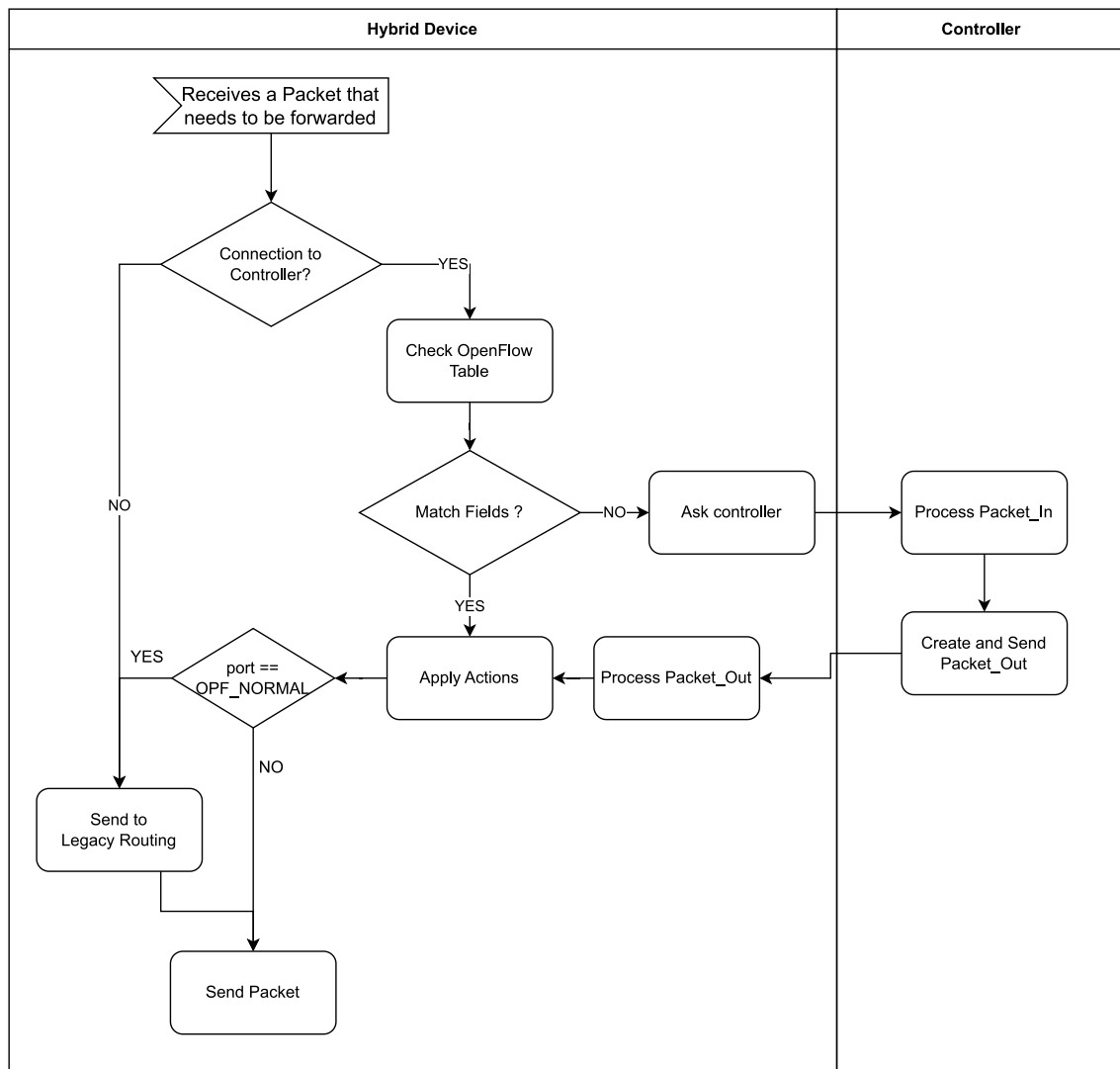


Fig. 2 Local routing scheme in Hybrid SDN device, where both the legacy routing mechanism and OpenFlow tables coexist, with the OpenFlow tables taking priority

Because the legacy routing mechanism is already running in the hybrid devices, no delay is incurred from inferring the network's topology when swapping mechanisms. When the connection to the controller is in-place, the OpenFlow table is consulted to determine if the packet matches any fields in the saved rules. If a match is found, the actions specified in that rule are applied.

When a packet does not match any OpenFlow rule stored in the SDN table (which determines the packet's path), the hybrid device consults the controller. Afterward, the controller responds with a set of actions that may include forwarding the packet to the legacy routing mechanism to be treated by that protocol (OPF_NORMAL) or to just forward it to the interface described by the rule.

Moreover, we do not have control over the legacy devices in our hybrid network, but we can use them to improve

reachability and performance. However, by not having control over these legacy devices, our routing algorithm (installed in the hybrid devices) has to cope with their forwarding rules and heterogeneity. By taking into account the state of each link and the nodes' capabilities and state, we can increase the performance and available services while utilizing the legacy devices, which increases the network range and may prolong its lifespan.

4.1 A modified Dijkstra's algorithm

Dijkstra's algorithm is a well-known and widely used method to find the shortest path in a graph, making it the base for many efficient routing protocols Makariye [29].

The standard Dijkstra algorithm can determine the optimal path by considering a weighted label, which can be

based on distance or any other variable. It is based on the assumption that labels are orderable and the subsequent labels, given by the sum, are monotonically non-decreasing. However, there are instances where certain metrics cannot be represented by a simple sum of parameters. For example, the path's throughput is equal to the lowest link's throughput on such path. Therefore, the standard sum for each variable cannot be applied.

Algorithm 1 describes the proposed algorithm, which is a multi-weighted, multi-variable Dijkstra-based algorithm. Notice that this algorithm is able to get the required information from the links and nodes through the proactive topology management approach described in Sect. 3.2.

Algorithm 1 A multi-weight modified Dijkstra algorithm

Require: Graph, source, targets

```

for each vertex  $n$  in Graph do
     $\text{dist}[n] := \text{infinity}$ 
     $\text{previous}[n] := \text{undefined}$ 
end for
 $\text{dist}[\text{source}] := 0$ 
 $Q := \text{the set of all nodes in Graph}$ 
while  $Q$  not empty do
     $u := \text{node in } Q \text{ with smallest } \text{dist}[\ ]$ 
    remove  $u$  from  $Q$ 
    if  $u$  in targets then
        remove  $u$  from targets
        if targets is empty then
            break
        end if
    end if
    for each neighbour  $v$  of  $u$  do
         $\text{pathcost}_{(\text{source},v)} := \text{Pathcost}(\text{pathcost}_{(\text{source},u)}, \alpha^e)$ 
         $\text{pathcost}_{(\text{source},v)} < \text{dist}[v]$  then
             $\text{dist}[v] := \text{pathcost}_{(\text{source},v)}$ 
             $\text{previous}[v] := u$ 
             $\text{paths}[v] = \text{paths}[u] + [v]$ 
        end if
    end for
end while
return previous, paths, dist

```

▷ continue if all the targets have not been reached

▷ where v has not yet been removed from Q **do**

▷ α^e represents the link values for edge

Instead of incrementing the route's cost with the $\alpha_{(u,v)}$ value for that edge, the PathCost function (described below) is used instead.

The network is modeled by an un-directed graph $G(V, E)$, where V is a set of nodes, and E is a set of links or edges. Each edge $e = (u, v)$ between nodes u and v represents a direct link with multiple information gathered.

An edge e is characterized by α^e

$$\alpha^e = \begin{bmatrix} \alpha_1^e \\ \alpha_2^e \\ \alpha_3^e \\ \alpha_4^e \\ \alpha_5^e \\ \alpha_6^e \\ \alpha_7^e \end{bmatrix} = \begin{bmatrix} -1 \times \text{DeliveryRatio}^e \\ \text{Delay}^e \\ -1 \times \text{Throughput}^e \\ -1 \times \text{Link_Stability}^e \\ P_{\text{Break}}^e \\ 1/\text{MaxTTL} \\ PI^v \end{bmatrix}, \quad (1)$$

where $\text{DeliveryRatio}^e \in [0, 1]$ represents the ratio of successfully delivered packets to the total number of packets sent within a specific time window. The $\text{Delay}^e \in [0, 1]$ is the average packet's delay (in seconds) in the same time window, given by

$$\text{Delay}^e = \frac{\min(\text{Delay}^e, \text{MaxDelayConstant})}{\text{MaxDelayConstant}}, \quad (2)$$

where the Delay^e is upper bounded by a MaxDelayConstant . Similarly, the $\text{Throughput}^e \in [0, 1]$ is a normalized interval upper bounded by a $\text{MaxThroughputConstant}$. It represents the number of bits delivered in the same time window. All of these describe common link properties for e . We use the network statistics for our time window, in 1 s intervals.

$\text{Link_Stability}^e \in [0, 1]$ is a ratio between the time (in seconds) that the link has consecutively been up, and a

heuristic constant representing a sufficient duration for a link to be considered stable (in our case, 60 s).

Link_Stability^e ∈ [0, 1] for edge *e* (from node *u* to node *v*) is given by

$$P_{\text{Break}}^e = \begin{cases} 1 & \text{if } J^v < J_{\min} \\ \frac{J^v + \text{SNR}^e}{2} & \text{if } J^v \in [J_{\min}, J_{\max}] \\ \text{SNR}^e & \text{if } J^v > J_{\max} \end{cases} \quad (3)$$

where J^v , in Joules, is the normalized energy value of the node *v*, and SNR^e , in dBm, is the normalized value of the signal to noise ratio. $J_{\min}$ and $J_{\max}$ are energy constants representing acceptable limits, where the energy wasted utilizing the node by sending and receiving traffic will not affect the survivability of the network.

The actual weight of the edge's hop in a path is given by the fraction $1/\text{MaxTTL}$, where MaxTTL represents the maximum Time-To-Live that a packet can have (the maximum amount of hops a path could have).

The PI^v , Possible Interference, represents a set with the routes and their Data Rates, that currently utilize node *v*. For example, if a node has two flows with 100kbps each, the PI^v value is 2×100 .

This Path Cost can be applied in any wireless network that can provide this type of parameters for every node. Again, this information is gathered through the approach in Sect. 3.2.

When a flow requests a path, the type of service is categorized into one of multiple categories (for example, Text, Video, and voice) depending on the Data Rate of each flow. Each category *c* has a specific $\theta_c = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7]$, which represents the importance of each α element for that specific category. By using different θ_c , the algorithm can dynamically adapt to several types of services (with their corresponding requirements).

Therefore, the $\text{PathCost}_{(\text{source}, \text{target})}$ for a route of category *c* comprised of a set of E' links is given by

$$\text{PathCost}_{(\text{source}, \text{target})} = \theta_c \cdot \begin{bmatrix} \max_{e \in E'}(\alpha_1^e) + 1 \\ \sum_{e \in E'}(\alpha_2^e) \\ \max_{e \in E'}(\alpha_3^e) + 1 \\ \max_{e \in E'}(\alpha_4^e) + 1 \\ \max_{e \in E'}(\alpha_5^e) \\ \sum_{e \in E'}(\alpha_6^e) \\ PI^{E'} / PI^G \end{bmatrix}. \quad (4)$$

All the values in the matrix (before the dot product with θ_c) are normalized (i.e. $e[0, 1]$), and therefore, no value can outweigh the importance of its corresponding metric, and the total cost of a path is always non-decreasing. With that, the dot product (also known as scalar product) gives a value with orderable and non-decreasing properties, as required by the standard Dijkstra algorithm. The smaller that value

is, the better that edge is. This is the same intuition that path planning algorithms use, trying to pick the shortest path to reach a destination. Here, our measure of *shortest* is not the distance, but all the metrics discussed above; a short edge means a good edge. This search is not restricting paths with only SDN nodes, and legacy nodes can also be considered irrespectively of their nature.

The entire path's delivery ratio bottleneck is given by the lowest Delivery Ratio of any link comprising the path, $\min_{e \in E'}(\text{DeliveryRatio}^e) = \max_{e \in E'}(-1 \times \text{DeliveryRatio}^e) = \max_{e \in E'}(\alpha_7^e)$. However, in the Dijkstra algorithm, the cost of a path must be non-decreasing (a path cannot become shorter by adding a new edge). Therefore, we add a +1 constant to keep the value in the range [0, 1]. The same logic applies to the path's throughput α_3 and link stability α_4 .

For the link break probability, the bottleneck is instead the highest value across all links comprising the path, and the best path is the one will the lower value for the break probability. Therefore, we just compute the maximum value $\max_{e \in E'}(P(\text{Break})^e) = \max_{e \in E'}(\alpha_5^e)$. The average packet delay will be given by the sum of the links' delays, $\sum_{e \in E'}(\text{Delay}^e) = \sum_{e \in E'}(\alpha_2^e)$. Similarly, the sum is performed for the number of hops α_6 in the path.

The last parameter represents the number of distinct routes and their Data Rates $PI^{E'}$ that can be affected by or can affect this new path E' , divided by PI^G , which represents the total amount of routes and their Data rates currently active in the graph. By dividing by the total number of routes in the Graph, the number becomes normalized.

4.2 Loop avoidance and path recalculation

In a hybrid network, we have both legacy and hybrid SDN devices, where a legacy routing mechanism is running in parallel with the SDN's forwarding tables. Therefore, there is a chance of having loops between opposing forwarding mechanisms. For example, if the controller orders node A to send the packet to node B, and node B, being a legacy device, consults its routing table and forwards the packet back to node A, a loop is formed.

The controller can detect these loops, if: (1) a specific flow already has a path, (2) the sender node is a legacy device, and the receiver node is on the path but would have already forwarded the packet. In this case, the path is deleted, and the flow is forwarded based on the legacy routing tables (OpenFlow NORMAL behaviour) until a new path is calculated.

If the legacy device sends the packet to a hybrid node outside of the calculated path, a new path is calculated with the receiver as the starting point, and the original path is updated using that path after the last registered legacy device (receiver from a hybrid device).

When the edges are removed from the graph, the affected paths are recalculated, in which higher Data Rates take precedence over lower Data Rate flows.

4.3 Optimization process

Optimization algorithms are commonly used to optimize a particular scenario. However, optimizing the Key Performance Indicators (KPIs) that are best suited for a given scenario is often more valuable and strategic than optimizing a specific scenario. By identifying the most suitable KPIs, we aim to obtain optimized parameters that are independent of the scenario and can be extrapolated to any scenario. This is because they optimize based on changing network conditions, rather than specific scenario conditions.

In a network there are several types of services, each with its own set of constraints and requirements. For instance, real-time services require a path with less jitter, while file services require a path with zero loss Chen et al [10]. To address this, our proposal is to optimize θ for each service to identify the most suitable KPIs. This will enable us to dynamically compute the best path for each service based on its unique requirements.

OpenBox is a versatile and efficient optimization system designed to solve black-box optimization (BBO) problems. OpenBox is particularly useful for problems where it is important to find the best set of parameters or hyperparameters to optimize a system or model. It is known for its flexibility, adaptability to various domains, and strong performance in finding optimal solutions in situations where traditional optimization methods might be less effective Li et al [27].

In a BBO, the characteristics of the optimization objective to get its global optimum cannot be used. Therefore, samples of parameters are created, tested, and evaluated until an optimal value is found. In our case, OpenBox utilizes Bayesian

Optimization. This algorithm automatically chooses which model to use between Gaussian Process and Probability Random Forest.

An initial set of samples is generated and evaluated on the problem environment. Based on a given objective function, the best set of samples is used to compute the next set of similar samples. This cycle is repeated until a stop condition is met (the maximum amount of cycles is reached or performance has stagnated).

Our problem is characterized by a hybrid network, with both legacy and hybrid devices, in which we want to increase the performance by choosing the best paths for specific flows, so the network lives longer and can reach higher levels of performance.

Network Simulator 3 is used to simulate such a scenario, where a θ value enters as input in the simulation and is used to calculate the weight of each path as described in Eq. 4. The overall simulation performance of using this θ value is gathered as output.

As shown in Fig. 3, θ samples are provided by OpenBox, which are then tested in Network Simulator 3, and the simulation performance values (delivery ratio and delay) are used by the optimization's objective function, f , to evaluate and compute the next most promising sample of values for a specific Data Rate.

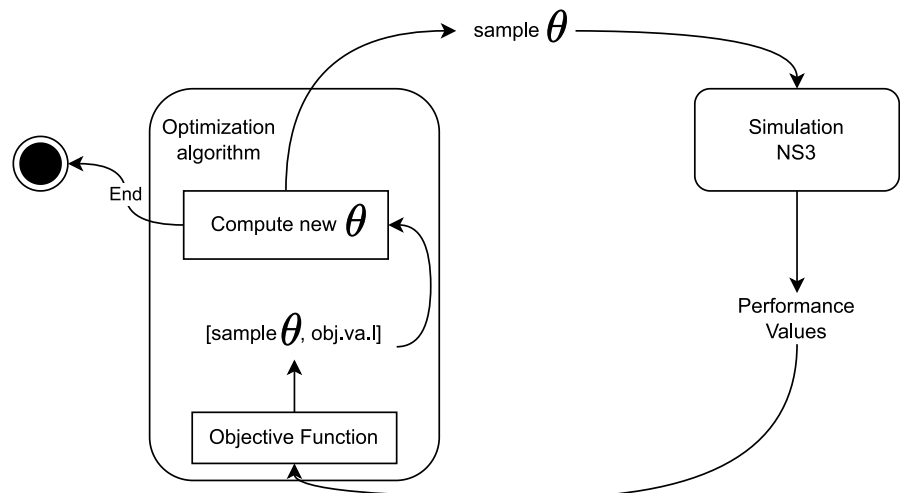
The following proposed objective function tries to increase the delivery ratio or at least reduce delay on the transmitted flows.

$$f = (10 \times (\text{delivery} + 1) - \text{delay}) \times (-1), \quad (5)$$

The values are multiplied by -1 because OpenBox treats the objective function as a loss function and minimizes it, and therefore f values are $\in [-1, 0]$.

Delivery and delay are values $\in [0, 1]$, and the best simulation is for increasing delivery ratio values, and

Fig. 3 Optimization mechanism



decreasing delay values, therefore we invert delay by multiplying it by (-1) .

Only unicast packets influence delivery ratio values, but even broadcast packets can influence delay values. Therefore, we can have an instance where we have a zero delivery ratio and a value for delay, which would result in a positive value for the objective function. In order to maintain f values $\in [-1, 0]$, we add 1 to the delivery ratio.

To imply that the delivery ratio takes precedence as the most important performance metric, we multiply it by 10 to increase its magnitude compared to the delay.

With the optimized values for θ , we can calculate the best paths for specific flows that will improve the performance, can increase the lifespan of the network, and allow for a better allocation of paths. This can increase the available capabilities in the legacy devices, as well as for the hybrid devices in the network.

5 Performance evaluation

To evaluate the proposed hybrid network and its routing approach, we design a hypothetical scenario in a rescue situation with multiple legacy and hybrid devices within three groups: one group for Local Government and Emergency Services, and two for rescue areas, as depicted in Fig. 4. These groups need to communicate amongst each other and within each other.

The first scenario comprises six legacy devices (in black), and nine hybrid software-defined capable devices (in red). It contains 15 flows between the legacy devices (all possible combinations), each with the same data rate.

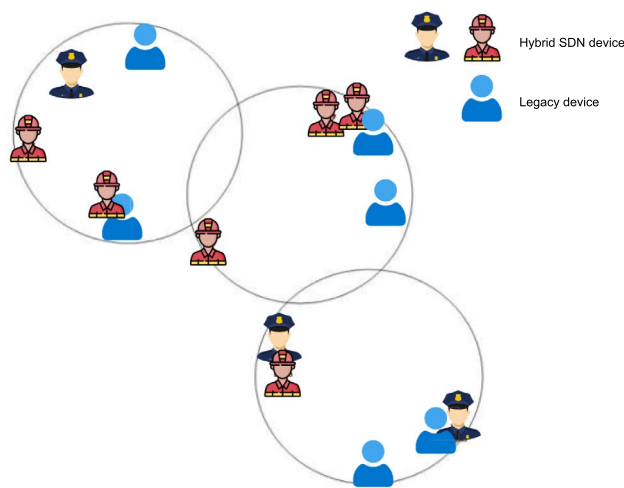


Fig. 4 Scenario of a disaster situation where teams of firefighters and police officers are deployed. This scenario is comprised of three groups, with six victims that represent the legacy devices, and nine rescuers that represent the hybrid SDN devices

These numbers of devices allow for a theoretical 100% connectivity for the flows at any given time, and the packet loss due to collisions or congestion is kept to a minimum with three groups. In addition, three larger scenarios will be considered to analyse how the variability of the number of nodes can influence the performance of the proposed approach.

The simulation is carried out in Network Simulator 3 with the module from Sousa et al [35], with a data plane with the following configurations:

- Wi-Fi Standard 802.11ax;
- Frequency band of 5GHz;
- Channel Width of 20MHz;
- IEEE 802.11ax configuration parameters: Modulation and Coding Scheme (MCS) Index of 3 and guard interval of 800 nanoseconds;

The mobility model is random, and controlled by a seed. Each group moves randomly inside a circle with a diameter of 180 m. Each group intersects with its nearest group, as depicted in Fig. 4. Each node moves with a uniform random speed $\in [0, 2]$ m/s towards a waypoint in its corresponding circle. The change of the speed of nodes will also be analysed in the overall scenario.

5.1 Optimization results

We consider the optimization of the network and its routing for five data rates: 100Kbps, 250Kbps, 450Kbps, 1Mbps, and 2Mbps.

Table 1 Optimized θ values for each data rate. Comparison with a Legacy only network

α	θ optimal values for different data rates				
	100 kbps	250 kbps	450 kbps	1 Mbps	2 Mbps
Delivery	0.184	0.130	0.211	0.141	0.145
Delay	0.078	0.168	0.116	0.168	0.186
Throughput	0.196	0.187	0.176	0.128	0.125
Link stability	0.173	0.138	0.102	0.120	0.140
Break probability	0.116	0.117	0.145	0.155	0.138
Number of hops	0.158	0.139	0.086	0.130	0.135
Possible interference	0.091	0.126	0.172	0.162	0.129
Hybrid net. abs(f)	930	926	847	499	269
Legacy net. abs(f)	895	832	684	384	223
Relative gain in f()	4%	11%	24%	30%	21%

Fig. 5 Best solution comparison with a Legacy network for 100Kbps

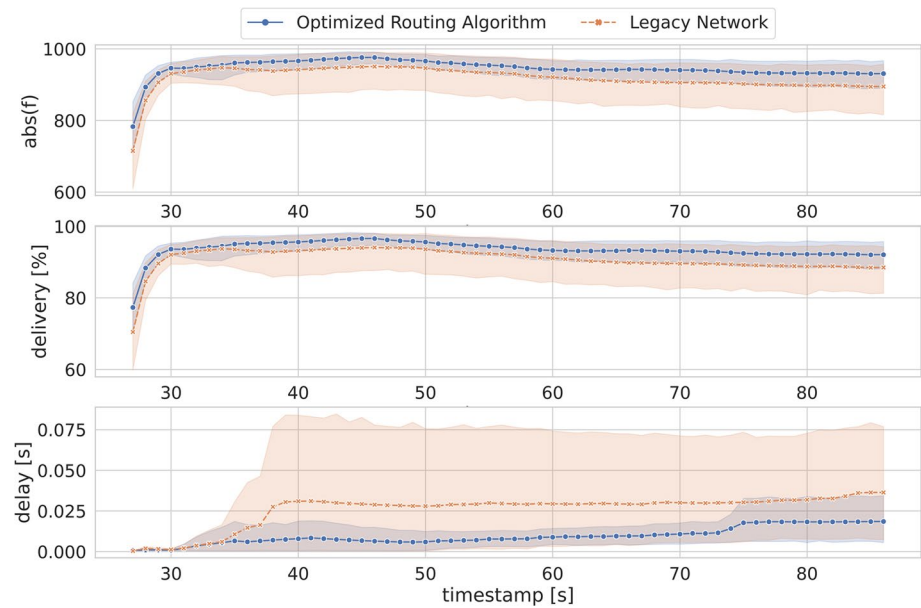
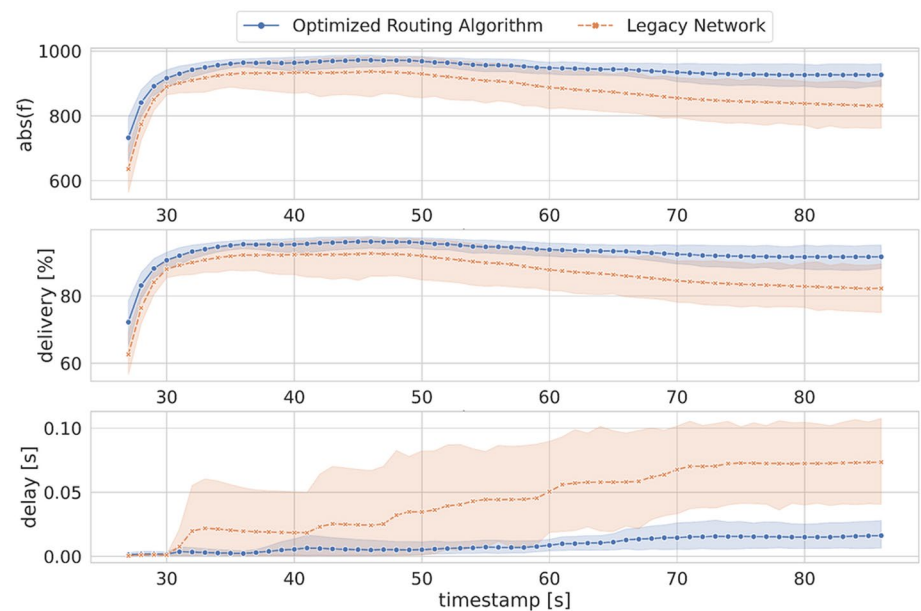


Fig. 6 Best solution comparison with Legacy network for 250Kbps



For each sample θ , a mean of 10 scenarios with different seeds is performed. In each scenario, 15 flows are sent for 60 consecutive seconds with specific data rates.

Table 1 represents the optimized θ values for each data rate and the relative gain when compared to the scenario with only Hop count as the path search metric.

The values represent the importance that α parameters have when performing a path search with the metric described in Sect. 4.1. Their sum should be 1 or near one (due to floating point constraints). Lower values mean that the α parameter does not impact the overall performance of the simulation. However, it can also be that a variety

of numbers were not tested, and the ones that were do not impact the performance. Higher values of θ means that the corresponding parameter is one of the important metrics for the data rate being tested.

The relative gain is calculated based on the f function for both the legacy network, and the hybrid network with the optimal θ values shown in Figs. 5, 6, 7, 8 and 9.

It is notable that either Delay or Delivery is one of the relevant parameters for each simulation, as they are metrics that make up the objective function.

As data rates increase, Delay becomes more important because when paths are congested, Delivery ratios decrease

Fig. 7 Best solution comparison with Legacy network for 450Kbps

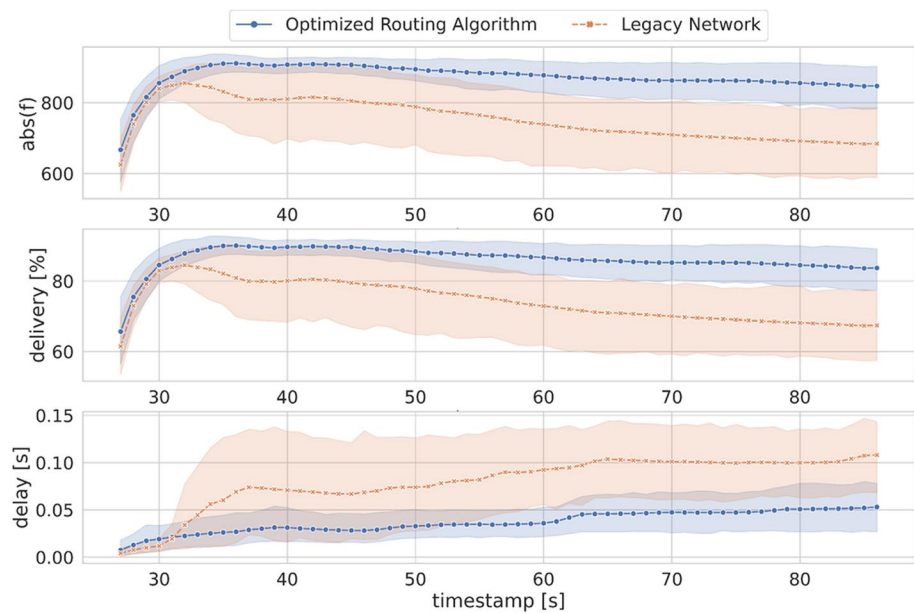
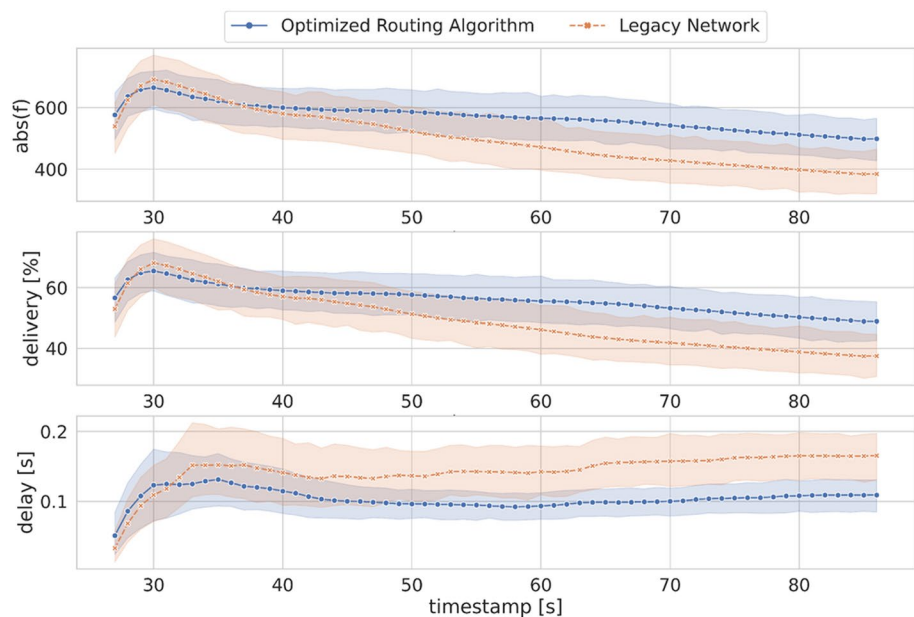


Fig. 8 Best solution comparison with Legacy network for 1Mbps



and become unchangeable and inconsistent due to packet loss. As a result, if there is no improvement in the Delivery ratio, its significance shifts to the Delay in accordance with the objective function f .

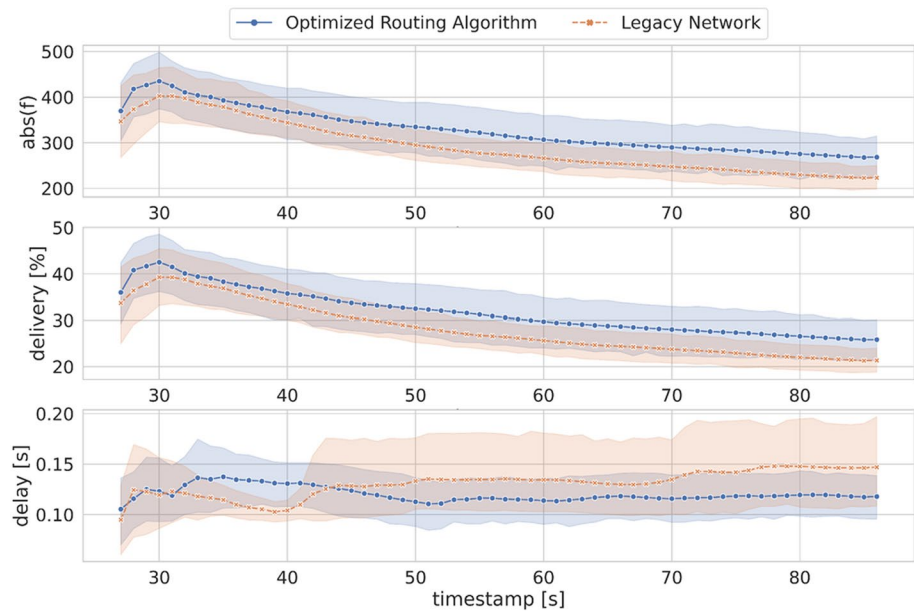
The Throughput weight decreases with higher data rates, which is unexpected. This can be due to the packet loss increase, which translates into noisy values in the Throughput, which does not accurately represent the state of the network.

For low data rates, Link Stability outweighs Break Probability, and the opposite happens for high data rates. Higher data rates mean more packets, which translates into more energy spending which has a high impact on the

Break Probability. Lower data rates have less impact on the Break Probability; therefore, Link Stability is instead more relevant.

When the Number of hops is highly relevant, the number of unique routes that can be affected (PI) is less important. However, the inverse also happens, and when one has an average importance, the same happens for the other metric. Combined, they have a high importance throughout every data rate.

When these optimal values are compared with a Legacy Network using a routing algorithm based only on hop count (for example OLSR), all data rates achieve better performance when multiple metrics are taken into account. Higher

Fig. 9 Best solution comparison with Legacy network for 2Mbps

data rates achieve higher relative gain (except the 2Mbps, because the packet loss is very high, so the gain cannot exceed those numbers).

Figures 5, 6, 7, 8 and 9 depict the performance of the network in terms of Delivery ratio, in %, and Delay, in seconds, for the specified injected data rates, from 100kbps up to 2Mbps, respectively. It also presents the objective function along the 60 s of experiment. All plots are averaged between 10 mobility seeds and also depict the 95% confidence interval. Plots in blue depict the simulations when the optimized θ is applied in the hybrid network; and the plots in orange depict the results when only using a legacy routing protocol.

As can be observed, the gain is mostly constant throughout the simulation, without outliers that could affect the average performance gain; and using only the hop count metric introduces greater variance across different mobility seeds. We can also observe that the Delivery Ratio of the simulation starts to decrease with higher data rates, even for the same scenarios. This can be explained by the increase in the packet loss due to congestion and collision in the network.

In Fig. 5, we can see that when the importance of the Number of hops is one of the most critical parameters, the paths chosen may not diverge from the legacy-only network. Therefore, less improvement can be achieved. In the legacy-only network, the Delay is higher than in the hybrid network due to utilizing the same path for multiple flows.

In Fig. 6, the overall Delay is kept to a minimum compared to the legacy-only network by using Delay as an essential parameter. We can also observe that the overall Delivery is maintained throughout the simulation, while the legacy-only network consistently loses more packets.

Table 2 Optimized θ values for different speed profiles, with 100Kbps. Comparison with a Legacy only network

α	θ Optimal values for different speeds		
	[0,2] m/s	[0,5] m/s	[0,10] m/s
Delivery	0.184	0.183	0.191
Delay	0.078	0.088	0.086
Throughput	0.196	0.118	0.081
Link stability	0.173	0.087	0.097
Break probability	0.116	0.211	0.200
Number of hops	0.158	0.187	0.199
Possible interference	0.091	0.123	0.147
Hybrid net. abs(f)	930	923	766
Legacy net. abs(f)	894	836	709
Relative gain in f()	4%	10%	8%

In Fig. 7, the difference between the two objective function's plots is increasingly higher throughout the performance. We can also observe that the network might be congested because the overall performance in the hybrid network is decreasing with time.

Figure 8 shows that in the beginning, when the paths are not congested, the legacy network with the Number of hops only metric outperforms the hybrid one. However, this is rapidly outperformed by the metric in the hybrid network.

5.2 Speed

Afterwards, we tested the same scenarios for the 100Kbps data rates, varying the speed at which all nodes move. Our

goal is to evaluate whether the optimal parameters would change across different speeds. Table 2 presents the results.

The relative gain is calculated based on the f function for both the legacy network and the hybrid network with the optimal θ values shown in Table 2.

As before, Link Stability remains the inverse of Break Probability, which in this case has a combined constant weight of 29%. A high Break Probability overshadows the Link Stability (in other words, the edge will break soon, so it does not matter if the link has been stable so far). The Throughput weight decreases with higher speeds because

more packets are lost, which creates a noisy metric that no longer accurately describes the state of the network.

The weights for the Possible Interference and Hop count increase, because longer routes will be more impacted by high speeds. Moreover, the Delivery Ratio and Delay weights remain constant.

This implies that we can further optimize our parameters if we had access to each node's speed. Obtaining these values falls beyond the scope of this work, but can be achieved if the proper hardware, like GPS, is integrated on the device and propagated into the controller.

5.3 Large scenarios

In this section, we test other scenarios with more groups, and therefore, more nodes in the simulation. Each group comprises two legacy devices, and 3 hybrid SDN devices. The amount of flows in each scenario is given by the combination of available legacy devices. Therefore, there are 15, 28, 45, and 66 flows in scenarios with 3 to 6 groups, respectively, all at a data rate of 250Kbps.

For each sample θ , a mean of 10 scenarios with different seeds is performed. In each scenario, each flow is sent for 60 consecutive seconds with 250Kbps.

Figures 10, 11 and 12 show the spatial configuration of each scenario. Table 3 presents the optimized θ values for each group and for each parameter.

As previously mentioned, either delay or delivery must be one of the relevant parameters which remains valid. These metrics are used to calculate the function f . If the delivery can significantly impact the overall performance, then θ is higher. Conversely, if the delivery has little to no effect, then the delay is the more significant factor. These changes are

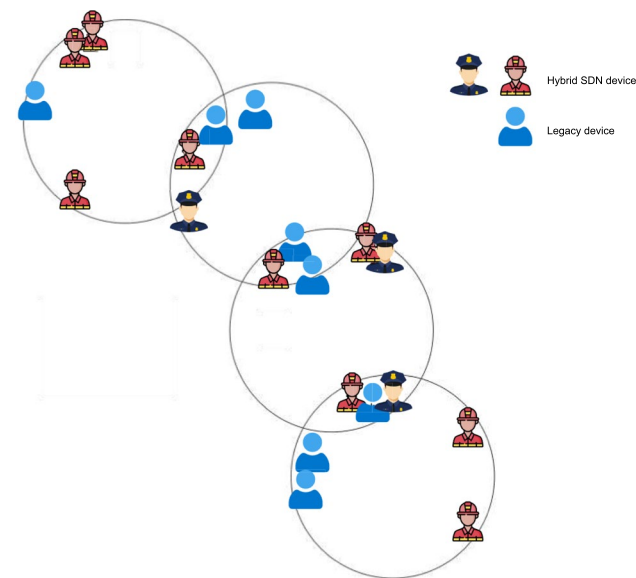


Fig. 10 Scenario with 4 groups, 8 legacy devices, and 12 Hybrid SDN devices

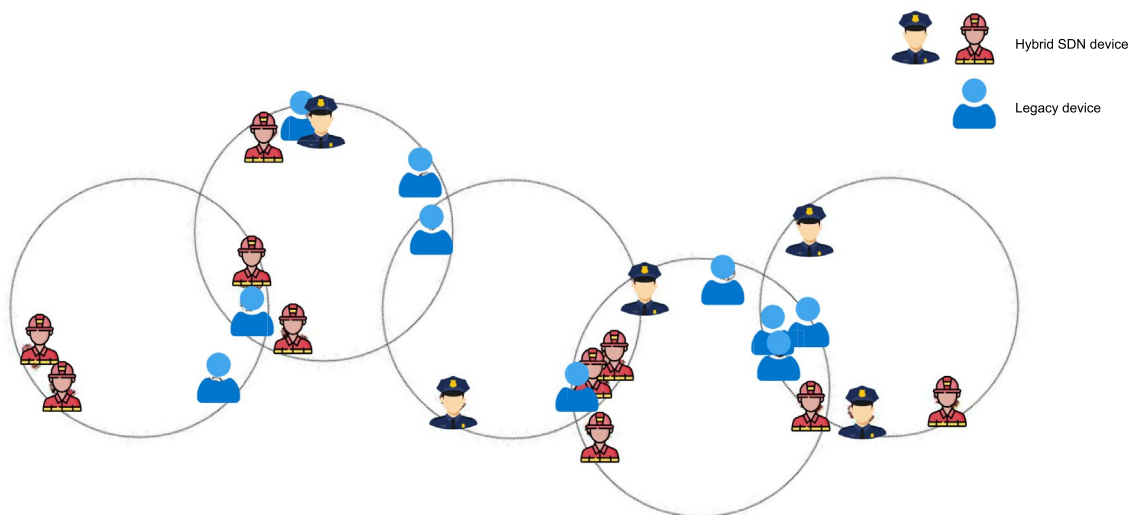


Fig. 11 Scenario with 5 groups, 10 legacy devices, and 15 Hybrid SDN devices

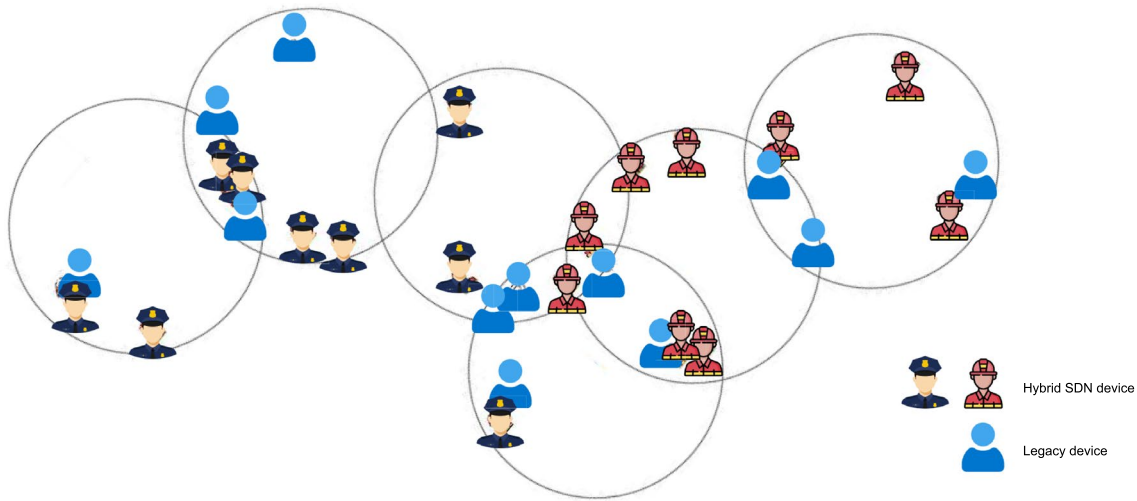


Fig. 12 Scenario with 6 groups, 12 legacy devices, and 18 Hybrid SDN devices

Table 3 Optimized θ values for different network sizes

α	θ optimal values for different number of groups			
	3 groups	4 groups	5 groups	6 groups
Delivery	0.130	0.149	0.205	0.110
Delay	0.168	0.190	0.129	0.162
Throughput	0.187	0.166	0.139	0.254
Link stability	0.138	0.115	0.125	0.134
Break probability	0.117	0.116	0.128	0.123
Number of hops	0.139	0.136	0.114	0.109
Possible interference	0.126	0.129	0.161	0.104

Table 4 Comparison with a non-optimized network (using θ values from the 250Kbps) and with a Legacy only network

	3 groups	4 groups	5 groups	6 groups
Legacy net. abs(f)	832	624	361	227
abs(f) using θ of 250 Kbps	–	641	409	262
relative gain in f()	–	3%	13%	15%
abs(f) using optimal θ values	926	700	445	288
relative gain in f()	11%	12%	23%	27%

more related to the scenario. Moreover, the delay remains one of the most important factors in more populated scenarios with this data rate.

When there are more flows that can utilize the same paths as other flows, the congestion and collision lead to an increase in the packet loss. Due to this, the throughput values do not change, making it an inaccurate parameter, and its

importance is shifted to other parameters. However, in the last scenario, this parameter becomes more important. The reason behind this is the change in spatial configuration of the groups which leads to the availability of more disjoint paths with less load.

As previously mentioned, Link Stability and Break Probability maintain a consistent sum of approximately 25%, in which the Link Stability has the same or higher importance. Therefore, we can conclude that these parameters are dependant of the data rate of each flow, and not to the mobility configuration of the scenario.

The importance of the Number of hops diminishes as other metrics become more important, but it is never a highly relevant parameter for this data rates.

As the number of flows between scenarios increases, it becomes increasingly important to prevent packet loss due to collision and interference (higher PI). However, in the last scenario, this parameter is less important because all possible paths are congested at the same level. Since there are too many flows in the network, the Possible Interference values do not vary. Consequently, the importance of this parameter decreases, and other more variable parameters become more significant.

Table 4 shows the relative gain compared to a legacy-only network. First, the optimal θ for the 250Kbps found in Table 1 for the scenario with three groups is used. We can see that, in all scenarios, the routing algorithm with the optimal θ can still perform better than the legacy-only network. However, higher performances can be achieved by optimizing θ , starting from the optimal θ of the first scenario.

As shown above, optimized parameters can bring benefits to different scenarios. Moreover, these values can be used in other scenarios with similar mobility characteristics and remain advantageous.

The correlation between Delivery and Delay depends on the load of the network: Delivery is more important when the available paths are not congested. However, either of them needs to be one of the most important metrics. The relation between Link Stability and Break Probability is influenced by the data rate and the speed at which the nodes move. For slower scenarios or with lower data rates, Link Stability is more important. Break Probability is more valuable for higher data rates when packet loss is not high. The Number of hops is only relevant at lower data rates and high speeds. The *PI* is dependent on the load of the network, being an important metric for higher data rates, and higher number of flows.

Finally, Throughput becomes unreliable when the packet loss is high, rendering it an unhelpful metric for higher data rates, highly populated, or high speed scenarios. Therefore, it remains to be seen if this metric should be combined with another metric that can account for the theoretical load of the network, so that the values are less erratic. Moreover, it remains to be seen if these correlations can generalize to new unseen scenarios. Nevertheless, they can serve as a starting point for the optimization algorithm to find an optimal value faster.

6 Conclusions and future work

In this article, we proposed an optimized adaptive routing approach in a network architecture that joins SDN-capable devices with legacy devices for emergency scenarios. A topology discovery mechanism is proposed to give the SDN controller an up-to-date view of such network, where the cost of the network is minimized (no overhead). This solution takes into consideration several aspects of the network when calculating a new heterogeneous path. One of them is the impact that a flow can have in the network, therefore minimizing its channel interference, and to a certain degree, balancing the load in the network's devices. In this work, the importance of the network and service parameters is optimized through machine learning for each flow's data rate, and then feed into the routing approach to determine the best hybrid legacy and SDN-based path.

The results show that flows with different data rates have distinct important parameters. Either delay or delivery ratio is a significant parameter, as they are metrics that make up the objective function. The throughput loses relevance with higher Data Rates, because it is an inaccurate parameter (due to congestion). The Link stability and Break Probability maintain the sum of importance in all scenarios. The Link stability is more relevant for lower Data Rates, while the Break Probability is more relevant for higher Data Rates.

Moreover, they maintain the sum with more flows with the same Data Rate, making this relationship independent of the scenario, but dependent on the Data Rate of each flow.

Moreover, the number of hops and possible interference combined have a high importance in every Data Rates. When the number of hops is highly relevant, the possible interference is less important. However, for scenarios with a higher number of nodes, the number of hops becomes increasingly less important while the possible interference becomes more important. The results also show that the nodes' speed influences the performance of the network, and so it should be a parameter taken into account in the QoS-metrics. However, it is not a value that can be readily computed by the network controller, and thus, its integration needs to consider an extended approach for GPS-based devices.

The proposed architecture and routing mechanism show that, when utilizing an adaptive routing metric with a Hybrid SDWN, the performance is increased or at least maintained when compared to Legacy-only networks. Our solution will be a baseline for a future QoS mechanism in a hybrid network, which allocates resources while increasing the network's performance.

The results here presented were obtained through simulation. Future work also includes evaluating our solution in a real testbed and comparing it against other network types (such as opportunistic networks) in emergency scenarios.

Acknowledgements This work was supported by the “Fundação para a Ciência e Tecnologia” (FCT) through the Grant SFRH/BD/143516/2019, and FCT/MEC through national funds under the project PTDC/EEI-COM/5284/2020 MH-SDVanet.

Author contributions All authors wrote and reviewed the manuscript. D.S. prepared the figures.

Funding Open access funding provided by FCTIFCCN (b-on).

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declares that they have no competing interests.

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